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Batch 15

Algorithm

1. Read the size `n`.
2. Read `n` integers into `nums[]`.
3. Pass the address of `nums[]` to `product_except_self()`.
4. Create `answer[n]`.
5. Set `prefix = 1`.
6. Traverse from left to right:
   - Store `prefix` in `answer[i]`.
   - Multiply `prefix` by `nums[i]`.
7. Set `suffix = 1`.
8. Traverse from right to left:
   - Multiply `answer[i]` by `suffix`.
   - Multiply `suffix` by `nums[i]`.
9. Display `answer[]` separated by commas.

Test Cases

Test	Input Array	Expected Output
1	`1 2 3 4`	`24, 12, 8, 6`
2	`-1 1 0 -3 3`	`0, 0, 9, 0, 0`
3	`2 3 4`	`12, 8, 6`
4	`5 2`	`2, 5`
5	`0 1 2 3`	`6, 0, 0, 0`
6	`0 0 2 3`	`0, 0, 0, 0`

Time complexity: `O(n)`
Division: Not used
Array passing: Done using pointer `int *nums`

C program]

```
#include <stdio.h>

void product_except_self(int *nums, int n)
```

```
{
    int answer[n];
    int prefix = 1;
    int suffix = 1;
    int i;

    /* Store product of all elements to the left */
    for (i = 0; i < n; i++)
    {
        answer[i] = prefix;
        prefix = prefix * *(nums + i);
    }

    /* Multiply by product of all elements to the right */
    for (i = n - 1; i >= 0; i--)
    {
        answer[i] = answer[i] * suffix;
        suffix = suffix * *(nums + i);
    }

    /* Display answer array */
    for (i = 0; i < n; i++)
    {
        printf("%d", answer[i]);

        if (i < n - 1)
            printf(", ");
    }

    printf("\n");
}

int main()
{
    int n, i;

    printf("Enter the size of the array: ");
    scanf("%d", &n);

    int nums[n];

    printf("Enter %d elements: ", n);

    for (i = 0; i < n; i++)
    {
        scanf("%d", &nums[i]);
    }

    product_except_self(nums, n);

    return 0;
}
```

Output

```
 Last login: Tue Oct 20 13:26:51 on console
[bhavya@Bhavyas-MacBook-Air ~ % nano day53.c
[bhavya@Bhavyas-MacBook-Air ~ % clang day53.c -o day53
[bhavya@Bhavyas-MacBook-Air ~ % ./day53
 Enter the size of the array: 4
 Enter 4 elements: 1 2 3 4
 24, 12, 8, 6
[bhavya@Bhavyas-MacBook-Air ~ % ./day53
 Enter the size of the array: 5
 Enter 5 elements: -1 1 0 -3 3
 0, 0, 9, 0, 0
 bhavya@Bhavyas-MacBook-Air ~ %
```